WORKED SOLUTION

STEP I 2014, Question 12

Probability and Statistics • Discrete Random Variables

STEP I 2014, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q12

The key idea

Squaring removes the modulus signs, so $\mathrm{E}(X^2)$ needs only $\mathrm{E}(S)$ and $\mathrm{E}(S^2)$ for the die score S, averaged over the two coin outcomes. The integer condition then turns into a divisibility question about $(k-1)^2$, which leaves very few single-digit candidates. Once k is known, X takes only five values, and the two-round game is a short enumeration of pairs.

Solution

Finding $\mathrm{E}(X^2)$

Let S be the number shown on the die, so

$$\mathrm{P}(S=1)=\tfrac{1}{2}, \qquad \mathrm{P}(S=2)=\tfrac{1}{3}, \qquad \mathrm{P}(S=3)=\tfrac{1}{6}.$$

Then

$$\mathrm{E}(S)=\tfrac{1}{2}+\tfrac{2}{3}+\tfrac{3}{6}=\tfrac{5}{3}, \qquad \mathrm{E}(S^2)=\tfrac{1}{2}+\tfrac{4}{3}+\tfrac{9}{6}=\tfrac{10}{3}.$$

The coin is fair and independent of the die. Since $|a|^2=a^2$, given heads $X^2=(kS-1)^2$, and given tails $X^2=(k-S)^2$. So

$$\begin{aligned}\mathrm{E}(X^2) &= \tfrac{1}{2}\,\mathrm{E}[(kS-1)^2]+\tfrac{1}{2}\,\mathrm{E}[(k-S)^2]\\ &= \tfrac{1}{2}[k^2\mathrm{E}(S^2)-2k\mathrm{E}(S)+1]+\tfrac{1}{2}[k^2-2k\mathrm{E}(S)+\mathrm{E}(S^2)]\\ &= \tfrac{1}{2}\left[\tfrac{10}{3}k^2-\tfrac{10}{3}k+1+k^2-\tfrac{10}{3}k+\tfrac{10}{3}\right]\\ &= \frac{13k^2-20k+13}{6}.\end{aligned}$$

On the other hand,

$$k+\frac{13(k-1)^2}{6}=\frac{6k+13k^2-26k+13}{6}=\frac{13k^2-20k+13}{6},$$

so $\mathrm{E}(X^2)=k+\frac{13}{6}(k-1)^2$, as required.

Finding k

Since k is an integer, $\mathrm{E}(X^2)$ is an integer exactly when 6 divides $13(k-1)^2$. As 13 is coprime to 6, this means $6 \mid (k-1)^2$, that is, both 2 and 3 divide $(k-1)^2$. Because 2 and 3 are prime, each must then divide $k-1$ itself, so $6 \mid k-1$. With $1 \le k \le 9$ this leaves $k = 1$ or $k = 7$.

> **Ask yourself** Both candidates make $\mathrm{E}(X^2)$ an integer. Which condition rules one out?

The condition on $\mathrm{E}(X)$ does. If $k = 1$, both rules give $X = |S-1| = S-1$, so $\mathrm{E}(X) = \frac{5}{3} - 1 = \frac{2}{3}$, which is not an integer.

If $k = 7$, then $7S - 1 > 0$ and $7 - S > 0$ for every face, so the modulus signs can be dropped:

$$\mathrm{E}(X) = \tfrac{1}{2}(7 \cdot \tfrac{5}{3} - 1) + \tfrac{1}{2}(7 - \tfrac{5}{3}) = \tfrac{16}{3} + \tfrac{8}{3} = 8,$$

and $\mathrm{E}(X^2) = 7 + \frac{13}{6} \cdot 36 = 85$. Both are positive integers, so $k = 7$.

The distribution of X

With $k = 7$, heads gives $X = 7S - 1 \in \{6, 13, 20\}$ and tails gives $X = 7 - S \in \{6, 5, 4\}$. Each (coin, die) combination has probability $\frac{1}{2}\mathrm{P}(S = s)$, and the value 6 arises both from heads with $S = 1$ and from tails with $S = 1$:

x	4	5	6	13	20
$\mathrm{P}(X = x)$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{12}$

The probabilities add to 1.

The two-round game

Let X_1, X_2 be the outcomes of the two rounds. They are independent, since each round uses a fresh roll and toss, and each has the distribution above. We need the pairs with sum above 25 and with sum exactly 25.

> **Ask yourself** How can we list the pairs with total over 25 without missing any?

Sort them by the larger value. If neither outcome is 20, the largest possible total is $13 + 13 = 26$, and $13 + 6 = 19$ is already too small, so $(13, 13)$ is the only such pair. If one outcome is 20, the other must exceed 5, so it is 6, 13 or 20; it equals 5 for a total of exactly 25. Hence

$$\begin{aligned}\mathrm{P}(X_1 + X_2 > 25) &= \mathrm{P}(13,13) + 2\,\mathrm{P}(20,6) + 2\,\mathrm{P}(20,13) + \mathrm{P}(20,20) \\ &= \tfrac{1}{36} + 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{6} + \tfrac{1}{144} \\ &= \frac{4 + 12 + 4 + 1}{144} = \frac{7}{48},\end{aligned}$$

where the factor 2 counts the two orders. Similarly

$$\mathrm{P}(X_1 + X_2 = 25) = 2\,\mathrm{P}(20,5) = 2 \cdot \tfrac{1}{12} \cdot \tfrac{1}{6} = \tfrac{1}{36}.$$

So the amount the gambler expects to be paid, in pounds, is

$$\frac{7w}{48} + \frac{1}{36}.$$

The game favours the casino when this is less than the £1 stake:

$$\frac{7w}{48} + \frac{1}{36} < 1 \iff \frac{7w}{48} < \frac{35}{36} \iff w < \frac{35 \cdot 48}{36 \cdot 7} = \frac{20}{3}.$$

As w is an integer, the largest possible value is $w = 6$. Since $\frac{20}{3}$ is not an integer, the answer is the same whether or not a fair game (expected payment exactly £1) is allowed.

Checks

- From the table, $\mathrm{E}(X) = \frac{4+10+36+26+20}{12} = 8$ and $\mathrm{E}(X^2) = \frac{16+50+216+338+400}{12} = 85$, matching the formulas.
- With $w = 6$ the expected payment is $\frac{7}{8} + \frac{1}{36} = \frac{65}{72} \approx$ £0.90, below the stake; with $w = 7$ it is $\frac{151}{144} \approx$ £1.05, above it.
- A computer simulation of 2×10^5 two-round games gave a total over 25 in about 14.5% of games ($\frac{7}{48} \approx 14.6\%$) and exactly 25 in about 2.8% ($\frac{1}{36} \approx 2.8\%$).

Common slips

- Listing $X = 6$ twice in the distribution, once for heads and once for tails, instead of combining them into $\mathrm{P}(X = 6) = \frac{1}{2}$.
- Forgetting the factor 2 for the two orders of an unequal pair such as $(20, 6)$.
- Treating "sum equals 25" as a win of £w or as a loss: the gambler just gets the stake back.

Beyond STEP

The calculation of $\mathrm{E}(X^2)$ is an instance of the law of total expectation, $\mathrm{E}(Y) = \sum_i \mathrm{P}(A_i)\,\mathrm{E}(Y \mid A_i)$, which first-year probability courses use constantly. The two-round game is about the distribution of a sum of independent random variables, a convolution, which generating functions handle neatly. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

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